In [2]:
```python
# Import necessary libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
import seaborn as sns
```

In [3]:
```python
# Load the dataset from the provided URL

url = 'https://github.com/akhandelwal8/globaleconomics/blob/main/hwk/hwk4a_tradedataBWA.csv?raw=true'
df = pd.read_csv(url, sep='\t')
```

1a

In [4]:
```python
total = df[['description', 'year', 'x']]
total = total.pivot(index='description', columns='year', values='x')
total['growth'] = (total[2023] - total[2000]) / total[2000] * 100

total
```

Out[4]:

year	2000	2023	growth
description			
Aircraft, spacecraft and parts thereof	4566759	4956454	8.533295
Albuminoidal substances; modified starches; glues; enzymes	28255	113896	303.100336
Aluminium and articles thereof	71525	4539418	6246.617267
Animal or vegetable fats and oils and their cleavage products; prepared animal fats; animal or vegetable waxes	2114015	1055469	-50.072776
Animal originated products; not elsewhere specified or included	2707	15564	474.953823
...	...	...	...
Wadding, felt and nonwovens, special yarns; twine, cordage, ropes and cables and articles thereof	35543	163635	360.386011
Wood and articles of wood; wood charcoal	423143	1127203	166.388195
Wool, fine or coarse animal hair; horsehair yarn and woven fabric	36380	3618	-90.054975
Works of art; collectors' pieces and antiques	5887	29707	404.620350
Zinc and articles thereof	5589	23037	312.184648

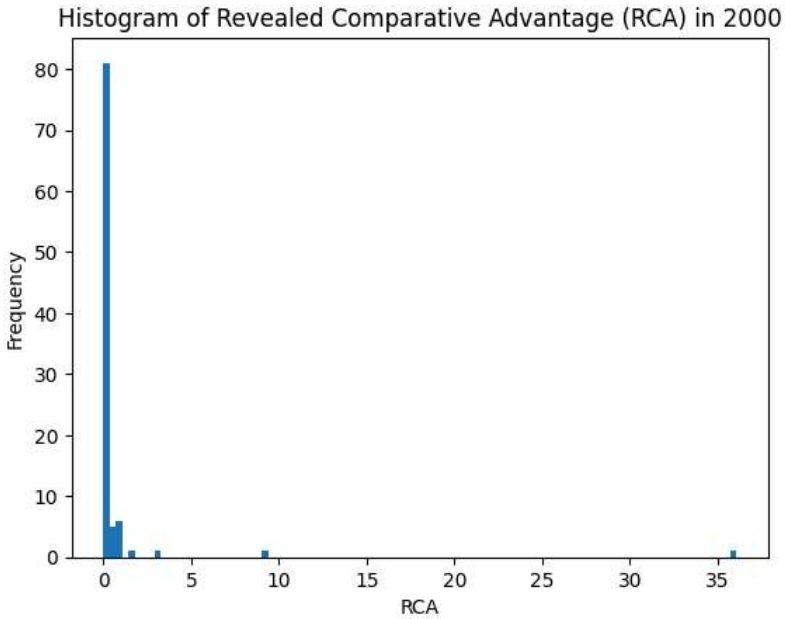
96 rows × 3 columns

1b

In [5]:
```python
df_00 = df[df['year'] == 2000].copy().reset_index(drop=True)
df_00['x_tot'] = df_00['x'].sum()
df_00['x_wd_tot'] = df_00['x_wd'].sum()
df_00['sh_hs2'] = df_00['x'] / df_00['x_wd']
```

```
df_00['sh_wd'] = df_00['x_tot'] / df_00['x_wd_tot']
df_00['rca'] = df_00['sh_hs2'] / df_00['sh_wd']
```

In [6]:
```
plt.hist(df_00['rca'], bins=100)
plt.xlabel('RCA')
plt.ylabel('Frequency')
plt.title('Histogram of Revealed Comparative Advantage (RCA) in 2000')
plt.show()
```



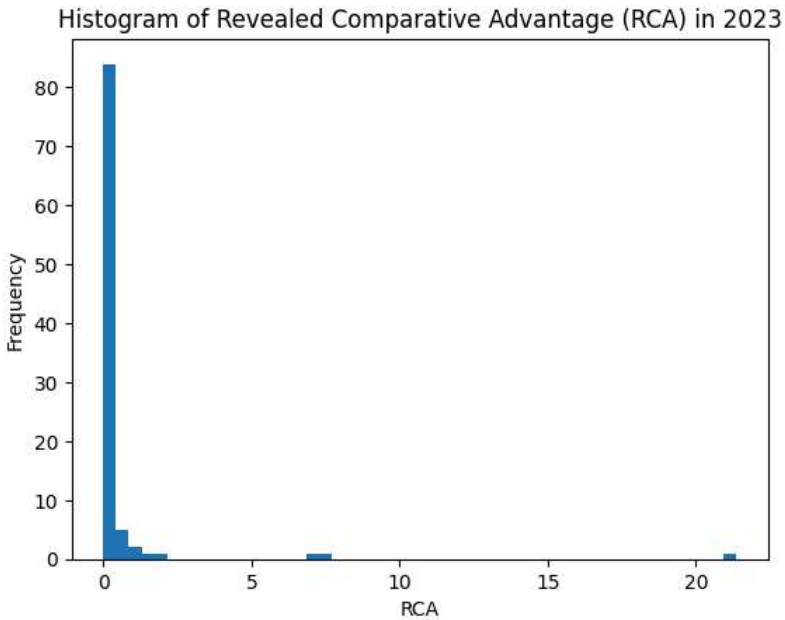
In [7]:
```
print(f"In 2000, Botswana had an RCA greater than or equal to 1 in {len(df_00[df_00['rca'] >= 1])} sectors.")
```

```
In 2000, Botswana had an RCA greater than or equal to 1 in 7 sectors.
```

1c

In [8]:
```
df_23 = df[df['year'] == 2023].copy().reset_index(drop=True)
df_23['x_tot'] = df_23['x'].sum()
df_23['x_wd_tot'] = df_23['x_wd'].sum()
df_23['sh_hs2'] = df_23['x'] / df_23['x_wd']
df_23['sh_wd'] = df_23['x_tot'] / df_23['x_wd_tot']
df_23['rca'] = df_23['sh_hs2'] / df_23['sh_wd']
```

In [9]:
```
plt.hist(df_23['rca'], bins=50)
plt.xlabel('RCA')
plt.ylabel('Frequency')
plt.title('Histogram of Revealed Comparative Advantage (RCA) in 2023')
plt.show()
```



In [10]:
```
len(df_23[df_23['rca'] >= 1])
```

Out[10]: 6

In 2023, Botswana had an RCA greater than or equal to 1 in 6 sectors.

1d

In [11]:
```
df_00[df_00['hs2'] == 71]
```

Out[11]:

	iso	hs2	description	year	x	x_wd	x_tot	x_wd_tot	sh_hs2	sh_wd	rca
70	BWA	71	Natural, cultured pearls; precious, semi-preci...	2000	2305254826	158544560128	2771865783	6876776872512	0.01454	0.000403	36.07284

In [12]:
```
df_23[df_23['hs2'] == 71]
```

Out[12]:

	iso	hs2	description	year	x	x_wd	x_tot	x_wd_tot	sh_hs2	sh_wd	rca
70	BWA	71	Natural, cultured pearls; precious, semi-preci...	2023	4357365455	974754676736	5500564093	26297198597376	0.00447	0.000209	21.371299

Botswana decreased its comparative advantage in HS71 (with the RCA decreasing from 36.073 to 21.371).

1e

In [13]:
```
df_00['rca_23'] = df_23['rca']
```

In [14]:
```
df_00['rca_change'] = df_00['rca_23'] - df_00['rca']
```

In [15]:
```
df_00.sort_values(by='rca_change', ascending=False).head(3)
```

Out[15]:

	iso	hs2	description	year	x	x_wd	x_tot	x_wd_tot	sh_hs2	sh_wd	rca	rca_23	rca_change
0	BWA	1	Animals; live	2000	235090	9799775232	2771865783	6876776872512	0.000024	0.000403	0.059516	7.301145	7.241630
25	BWA	26	Ores, slag and ash	2000	137671	26425796608	2771865783	6876776872512	0.000005	0.000403	0.012925	7.197501	7.184576
76	BWA	78	Lead and articles thereof	2000	9434	1736383232	2771865783	6876776872512	0.000005	0.000403	0.013479	1.659831	1.646352

The top 3 industry sectors in which Botswana's comparative advantage increased, based on absolute increases in the RCA, are (1) Live Animals, HS1, (2) Ores, Slag, and Ash, HS26, and (3) Lead and Articles Thereof, HS78.

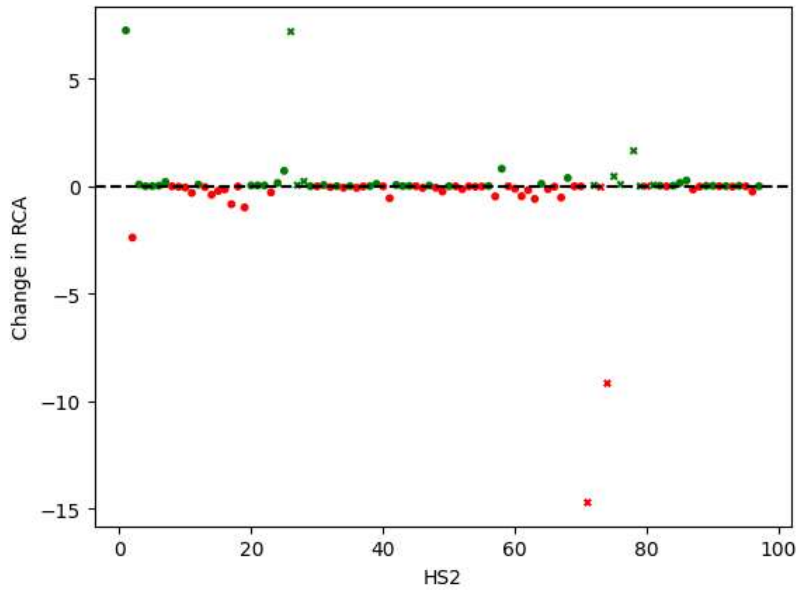
1f

In [16]:
```
minerals = [26,27,28,71,72,73,74,75,76,77,78,79,80,81]

mineralData = df_00[df_00['hs2'].isin(minerals)]
nonMineralData = df_00[np.invert(df_00['hs2'].isin(minerals))]

colors1 = ['r' if RCAdelta < 0 else 'g' for RCAdelta in mineralData['rca_change']]
colors2 = ['r' if RCAdelta < 0 else 'g' for RCAdelta in nonMineralData['rca_change']]

plt.figure()
plt.scatter(mineralData['hs2'],mineralData['rca_change'],10,marker='x',c=colors1)
plt.scatter(nonMineralData['hs2'],nonMineralData['rca_change'],10,marker='o',c=colors2)
plt.axhline(0,color='k',linestyle='--')
plt.xlabel("HS2")
plt.ylabel("Change in RCA")
plt.show()
```



Per the above scatterplot, most sectors showed very little change in the RCA in these 25 years. If anything, most non-resource sectors (marked as circles), showed marginal declines in the RCA (shown in red). However, there are a handful of sectors in which the increase in RCA is more noticeable. Of these, only about half are non-resource/non-mineral sectors. The other half of sectors in which Botswana's comparative advantage markedly increased were in mineral sectors (although the two largest declines were also in mining industries). Overall, the data shows mixed results in Botswana's comparative advantage gains in non-resource sectors. While a handful showed major improvement, the majority stayed mostly the same.

2)

In [17]:
```
# Load the dataset from the provided URL

url = 'https://github.com/akhandelwal8/globaleconomics/blob/main/hwk/hwk4b_tradedataZAF.csv?raw=true'
df1 = pd.read_csv(url, sep='\t')
```

2a)

In [18]:
```
world = df1.groupby('hs2')['xusa'].sum().reset_index(drop=True)
world_tot = df1['xusa'].sum()
```

In [19]:
```
df1 = df1[df1['iso'] == 'ZAF'].copy().reset_index(drop=True)
```

In [20]:
```
df1['x_wd'] = world
df1['x_wd_tot'] = world_tot
```

In [21]:
```
df1['x_tot'] = df1['xusa'].sum()
```

In [22]:
```
df1['sh_hs2'] = df1['xusa'] / df1['x_tot']
df1['sh_wd'] = df1['x_wd'] / df1['x_wd_tot']
df1['rca'] = df1['sh_hs2'] / df1['sh_wd']
```

2b)

In [23]:
```
df1.sort_values(by='rca', ascending=False).head(5)
```

Out[23]:

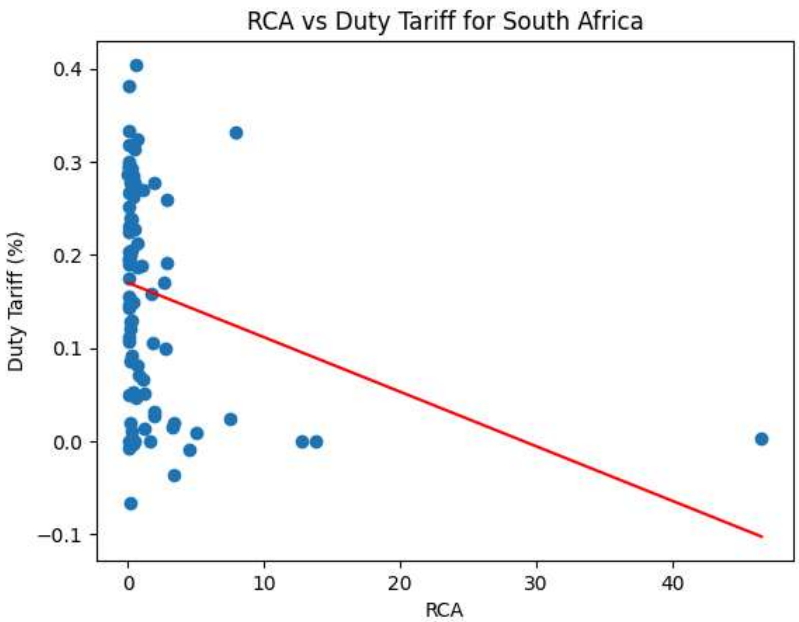
	year	iso	hs2	description	xusa	dtariff_zaf	x_wd	x_wd_tot	x_tot	sh_hs2	sh_wd	rca
21	2023	ZAF	26	ORES, SLAG AND ASH	3.032000e+08	0.002482	2.935200e+09	3.748076e+12	8.312755e+09	0.036474	0.000783	46.575214
61	2023	ZAF	71	NATURAL OR CULTURED PEARLS, PRECIOUS OR SEMI-P...	2.954000e+09	0.000055	9.688400e+10	3.748076e+12	8.312755e+09	0.355358	0.025849	13.747439
65	2023	ZAF	75	NICKEL AND ARTICLES THEREOF	1.527000e+08	0.000000	5.398700e+09	3.748076e+12	8.312755e+09	0.018369	0.001440	12.753025
66	2023	ZAF	76	ALUMINIUM AND ARTICLES THEREOF	5.957000e+08	0.331613	3.411570e+10	3.748076e+12	8.312755e+09	0.071661	0.009102	7.872935
35	2023	ZAF	41	RAW HIDES AND SKINS (OTHER THAN FURSKINS) AND ...	1.426613e+07	0.023800	8.599661e+08	3.748076e+12	8.312755e+09	0.001716	0.000229	7.479771

Top five sectors are HS26, HS71, HS75, HS76, and HS41.

2c)

In [24]:
```
plt.scatter(df1['rca'], df1['dtariff_zaf'])
plt.plot(np.unique(df1['rca']),
         np.poly1d(np.polyfit(df1['rca'], df1['dtariff_zaf'], 1))(np.unique(df1['rca'])), color='red')
plt.xlabel('RCA')
plt.ylabel('Duty Tariff (%)')
plt.title('RCA vs Duty Tariff for South Africa')

plt.show()
```



As daily tariffs on South Africa increase, the nation's RCA seems to decrease. This is due to increased frictions in international trade between the US and South Africa.

2d)

New tariffs will likely lead to a lesser concentration in SA's exports to the US, as SA will start to prioritize other countries by having a higher RCA with them.